

Aditya Kumar Singh

+91 9572234475 | adityasinghstuff@gmail.com | [linkedin.com/in/adityaonrepeat](https://www.linkedin.com/in/adityaonrepeat) | github.com/adityaonrepeat | adityaonrepeat.tech | [Loom Demo](#)

EDUCATION

National Institute of Technology, Agartala

Agartala, India

Bachelor of Technology CGPA: 8.17

2023 - 2027

EXPERIENCE

Jarurat Care Foundation

Jul 2026 - Present

Full Stack Developer Intern

Remote

- Engineered an **AI-assisted healthcare portal** (Next.js, TypeScript, Supabase), deploying **Zod-validated route handlers** and secure server-side APIs to support 3 critical patient workflows in production.
- Automated **lead generation** by engineering a **Python-based web scraping script** to extract healthcare-professional contact information, structuring the data into automated **CSV exports** that fed targeted outbound email campaigns.
- Eliminated recurring third-party subscription costs** by migrating campaign email infrastructure from **Bouncify** to **Amazon SES**, building an automated **AWS SNS** pipeline for bounce, complaint, and unsubscribe handling.
- Mitigated a critical security vulnerability** in the SES/SNS webhook via **AWS signature verification**, preventing unauthenticated manipulation of the global email suppression list.
- Integrated a **Gemini-powered healthcare chatbot** into the patient portal, engineering server-side guardrails and safety-constrained prompting for secure, structured LLM responses.

PROJECTS

QueryLoop | *Python, LangGraph, FastAPI, Chroma, PostgreSQL, Plotly, Next.js, TypeScript*

[Live](#) | [GitHub](#)

- Built a self-correcting **natural-language-to-SQL** agent as a LangGraph state machine over a **100,000+ order, 8-table** e-commerce database, feeding live PostgreSQL errors back to the model to repair failed queries across up to **3 attempts**, sandboxed behind a read-only, SELECT-only role.
- Achieved **85.7% first-pass execution accuracy** and **100% query execution success** on a self-designed 23-case benchmark by comparing generated result sets against golden answers using row-level, order- and float-tolerant evaluation.
- Grounded generation with a schema-RAG layer (Chroma + local MiniLM embeddings) retrieving the **top-4 relevant tables** per question and encoding documented domain gotchas, guarding against common join and double-count errors.
- Streamed the agent's full reasoning trace (retrieval → SQL → repairs → answer) to a Next.js/TypeScript UI over **Server-Sent Events**, with run persistence and live Plotly table/chart rendering.

Bloom | *Next.js, Socket.IO, PostgreSQL, Prisma, Redis, Gemini AI, ZegoCloud, Recharts*

[Live](#) | [GitHub](#) | [Demo](#) | [Blog](#)

- Built a **Redis matchmaking engine** across 4 emotion queues with same-emotion-first pairing, cross-emotion fallback, previous-peer exclusion, and atomic re-queuing to prevent double-matching; achieved **<1s pairing**, load-tested to **50+ concurrent connections**.
- Architected a **2-service deployment** (Next.js/Vercel + Socket.IO/Render) powering exclusive 1:1 ZegoCloud video rooms with cooldown-guarded skip, self-match prevention, and a peer-report pipeline auto-blocking after 10 reports; diagnosed and eliminated login redirects caused by Render/Neon cold starts via a scheduled keep-alive endpoint.
- Built a **streaming AI therapist** over SSE with token-level streaming, per-session memory, retry-on-overload for Gemini failures, and **sub-second time-to-first-token**, driven by a PHQ-9-inspired emotional scoring model powering 4-state matchmaking and 7/30/90-day mood history.

CampusKart | *Next.js, Socket.IO, PostgreSQL, Prisma, Upstash Redis, Sightengine AI*

[Live](#) | [GitHub](#) | [Demo](#)

- Built a college-exclusive marketplace (**100+ users, 5+ institutions**) with P2P listings, stores, and real-time chat; enforced hard per-institution data isolation via a **3-tier RBAC** (user / college-scoped moderator / global admin) validated on every server action, and gated every listing behind Sightengine AI image/text moderation with Upstash Redis rate limiting (5 listings/hr).
- Engineered real-time messaging, debounced typing indicators (one event per 1.5s idle burst, not per keystroke), and presence across **50+ concurrent sockets** on an independent Socket.IO server, deployed separately from the serverless Next.js app (Render + Vercel) since serverless functions can't hold persistent WebSocket connections.
- Eliminated N+1 unread-count queries with a **single grouped SQL join**, added composite indexes matched to real query shape (browse-feed filters, inbox sort, thread pagination), and shipped composite-cursor (createdAt, id) pagination supporting **200+ listings** with no dropped or duplicated pages.

ScamShield | *TensorFlow.js, Machine Learning, On-device AI, Chrome Extension (Manifest V3)*

[GitHub](#) | [Demo](#)

- Built a **privacy-first, zero-server-call** Chrome extension for on-device phishing detection, training a TensorFlow.js classifier on **291K+ URLs** to **88.47% test accuracy** across 16 engineered, min-max scaled features.
- Designed a hybrid risk engine blending heuristics (**30%**) and ML (**70%**) with independent escalation triggers (≥ 70 heuristic score or $\geq 90\%$ ML confidence), prioritizing low false-negative rates for high-confidence phishing threats.
- Engineered **homograph and IDN spoofing detection** via Punycode prefix checks and Cyrillic confusable-character matching, catching attacks that bypass standard URL pattern filters.

TECHNICAL SKILLS

- Languages:** Python, TypeScript, JavaScript, SQL, C++, C
- Programming Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming (OOP)
- AI & ML:** LangGraph, LangChain, RAG (Retrieval-Augmented Generation), Vector Search (Chroma), Embeddings, LLM Evaluation & Benchmarking, Prompt Engineering, Gemini API, TensorFlow.js, On-device Inference
- Backend & APIs:** FastAPI, Node.js, Socket.IO, REST, Server-Sent Events (SSE), Prisma, Better Auth
- Databases:** PostgreSQL, MongoDB, Redis, Supabase, Chroma
- Frontend:** React, Next.js, Tailwind CSS, Zustand, Redux, TanStack Query, Framer Motion, Web Extension Framework
- Tools & DevOps:** AWS (EC2, Amazon SES, SNS), Docker, Git, GitHub Actions, Postman, Bash, Vercel, Render

ACHIEVEMENTS

- Finalist: **Top 10** at GDG IIIT Agartala Hackathon; **Top 50** out of 578 at Hackshodh 2026, CSJMU Kanpur.
- Solved **500+ DSA problems** across [LeetCode](#) & [Codeforces](#).